



Marine Biology

海洋生物学



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Review of last week

- 1. Definition of Marine Biology**
- 2. History of Marine Biology**
- 3. Modern Marine Biology**
- 4. Scientific Method (Hypothesis, test & way of thinking)**
- 5. General Geography of the Ocean Basins**
- 6. Structure of the Earth**
- 7. Continental Drift**
- 8. Theory of Plate Tectonics**
- 9. Continental Margins, Deep-Ocean Basin, Hydrothermal vents**

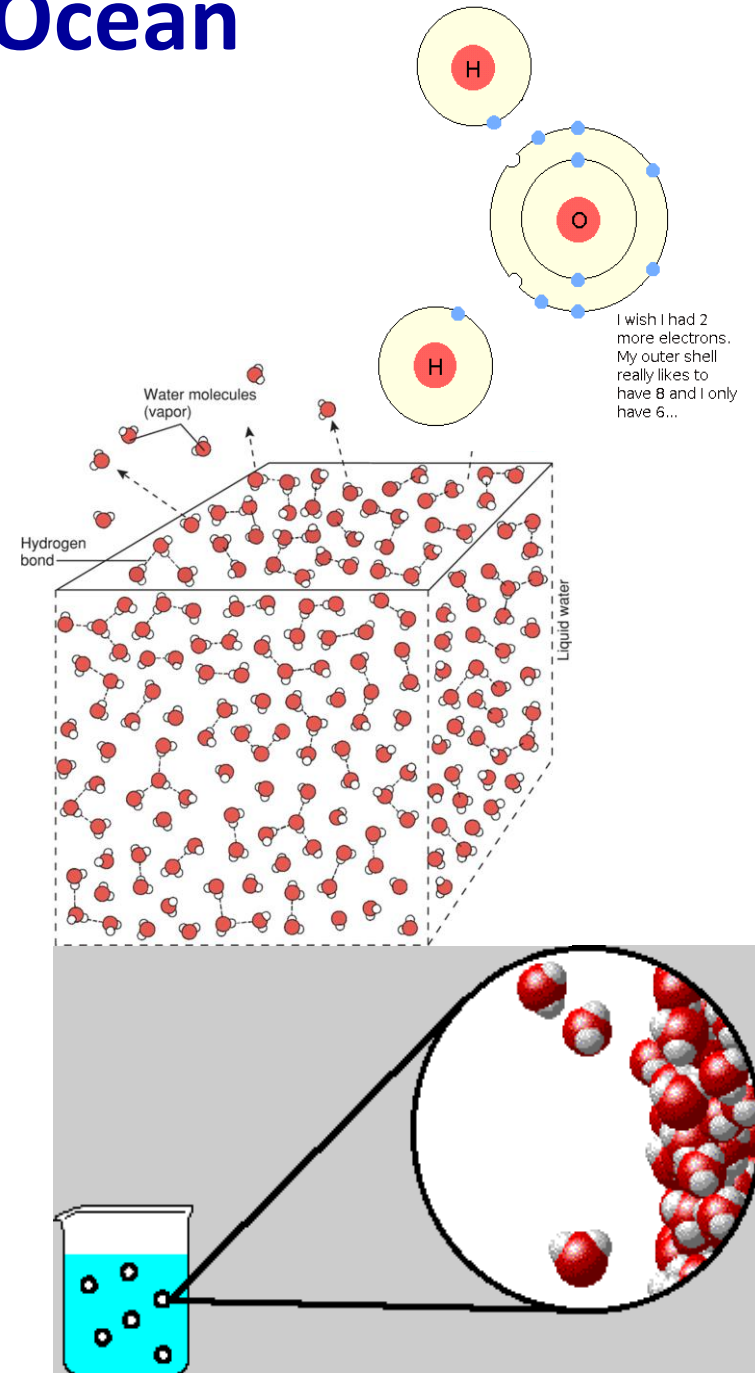


Chemical and Physical Features of Seawater and the World Ocean

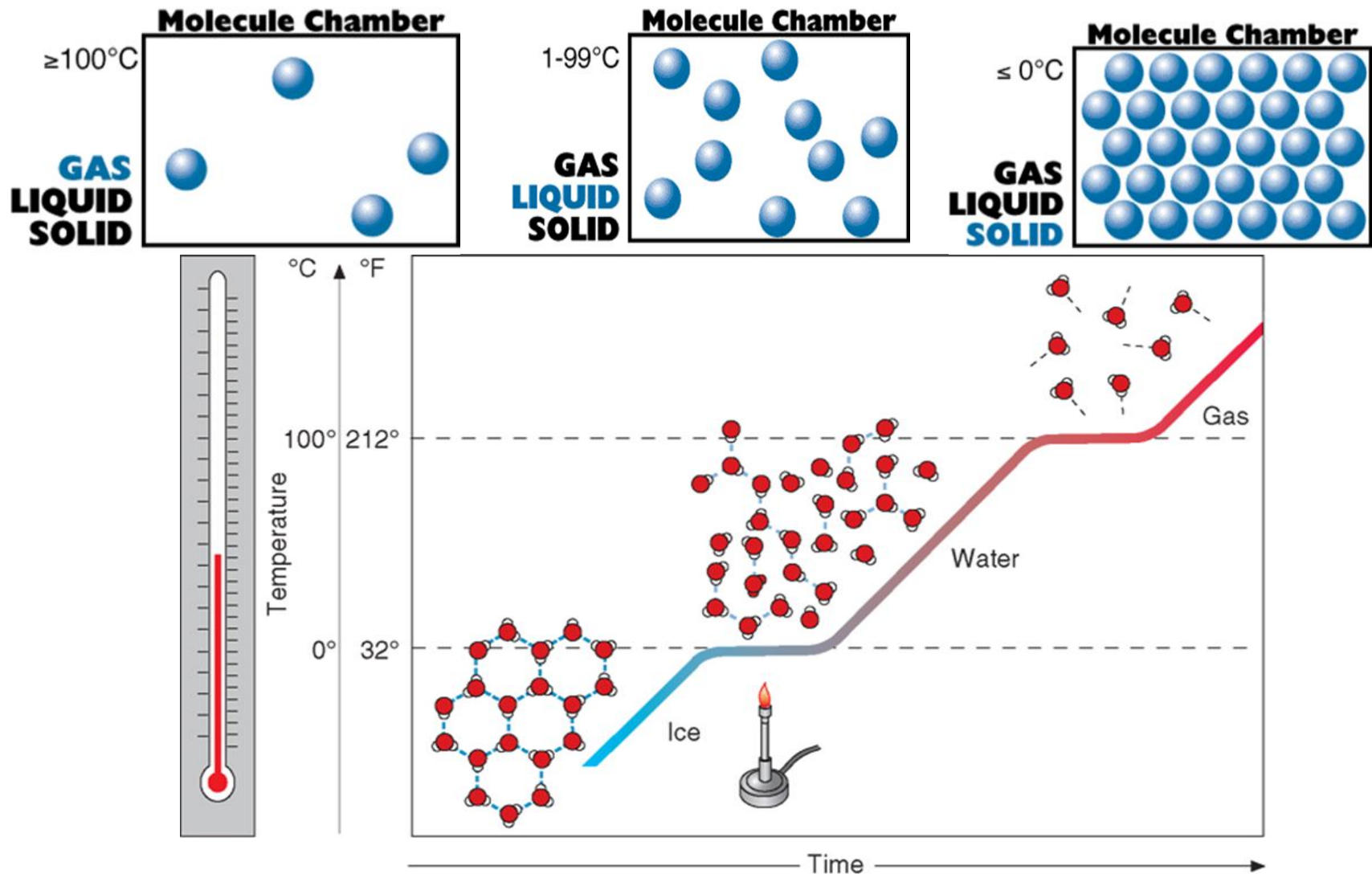
Part 1: The Waters of The Ocean

1. The Three States of Water

- In water (H_2O) molecule, **two atoms of hydrogen (H)** and **one atom of oxygen (O)** bonded chemically together
- Each water molecule has slight positive (H^+) and negative (O^{2-}) electrical charges, which make water molecules to be attracted to one another, known as **hydrogen bonding (HB)**
- HBs help to keep water molecules as a cohesive group (Liquid form) at most temperatures found on earth



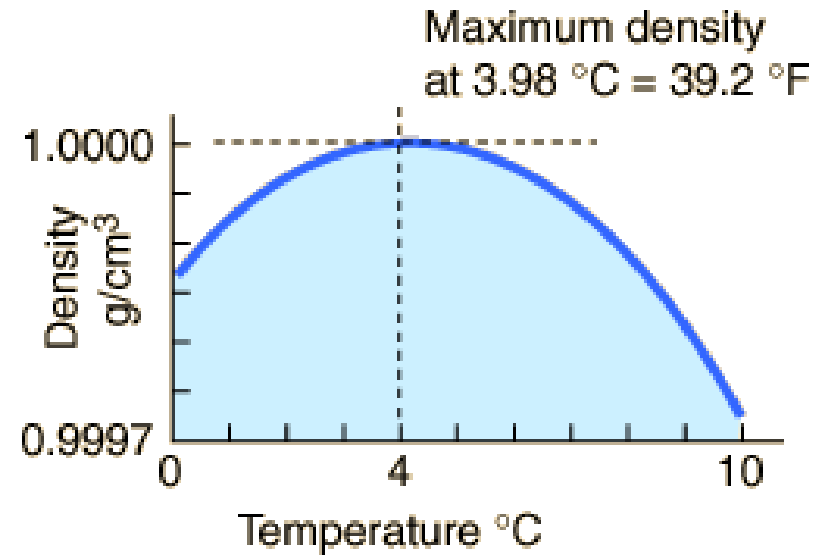
- Water is the only substance on earth that exists in all three states **naturally**, HB permits water to change phase when energy changed



2. Heat and Water

- With **lower temperatures** ($> 4\text{ }^{\circ}\text{C}$), water molecules move **closer** to one another and have **more density**

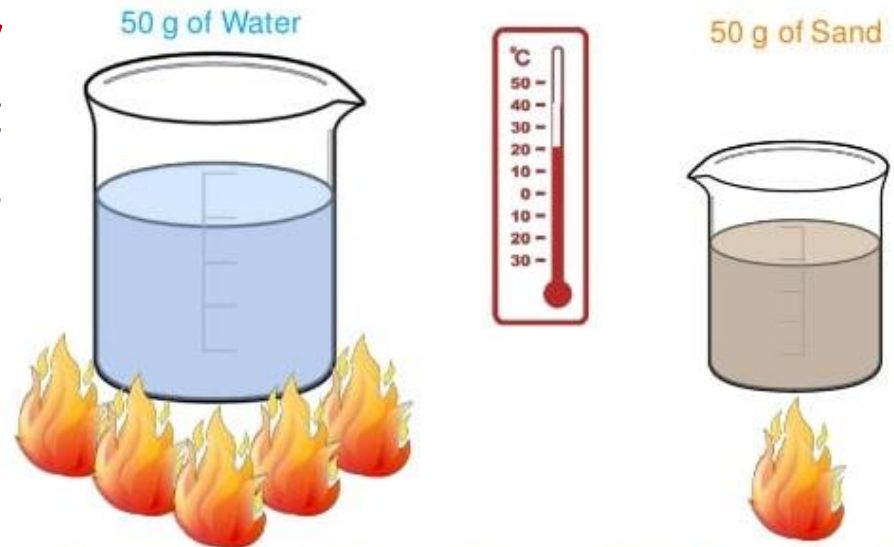
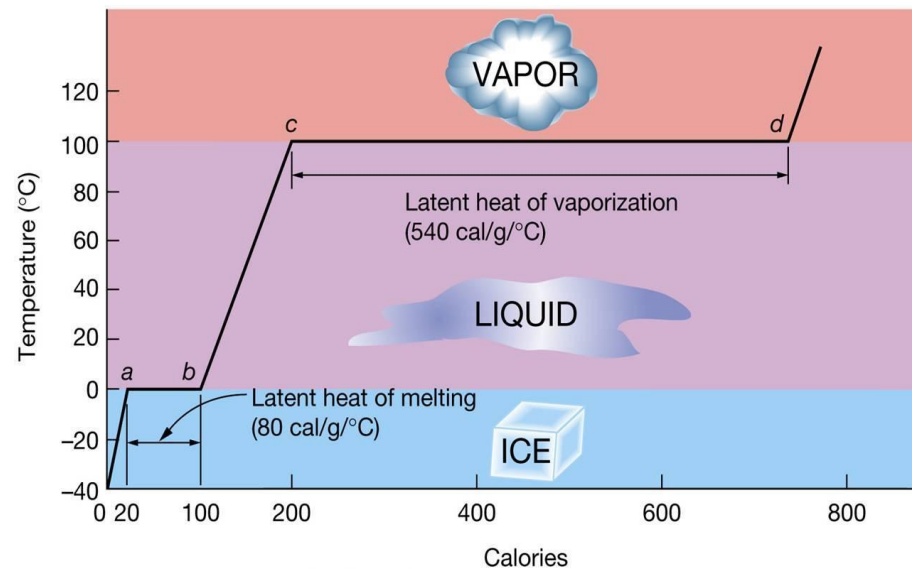
❑ Cold water can sink underneath the warmer water, hold **more oxygen** than the same volume of water to benefit organisms living at the ocean floor



- The situation changes when the water gets cold enough to freeze ($< 4\text{ }^{\circ}\text{C}$). Ice is less dense than liquid water due to all the air trapped in it, therefore, it floats

❑ Since ice floats, the floating ice creates a barrier between the air temperature, keeps water below the ice and organisms in the water from freezing, provides space for organisms

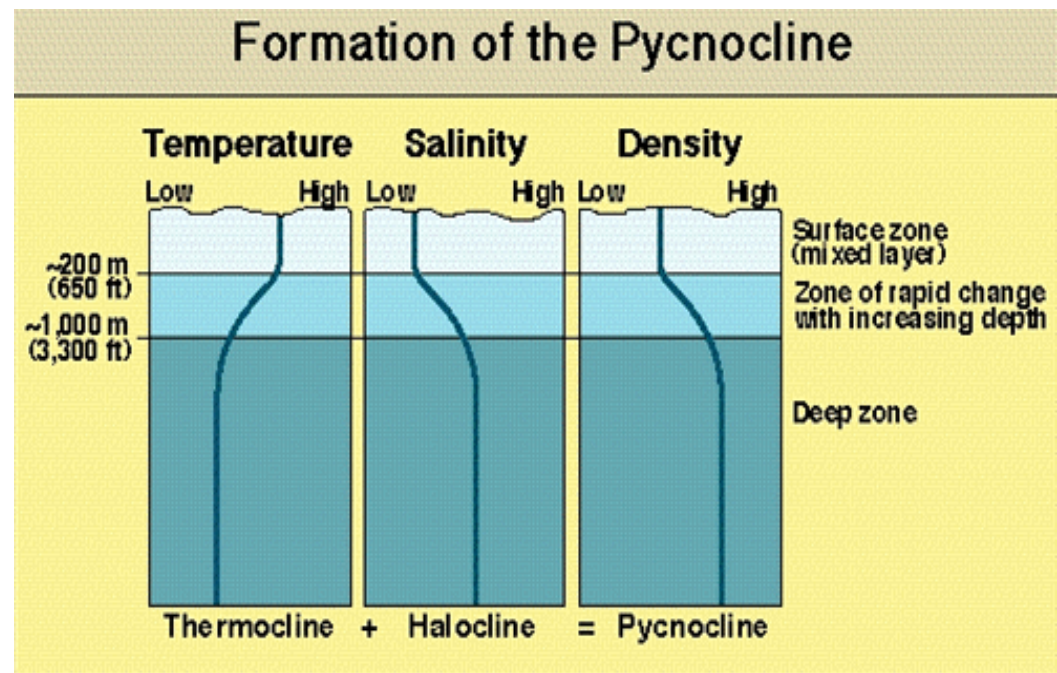
- Water has **high latent heat of melting** (融化潜热): the amount of heat required to melt
- Water has **high latent heat of evaporation** (蒸发潜热): the amount of heat required for a substance to evaporate
- Water has **high heat capacity** (热容): the amount of heat required to raise a substance's temperature by a given amount
 - ❑ Water absorbs a great deal of heat before its temperature rises
 - ❑ High heat capacity keep water temperature with a relative narrow range which is great for marine organisms



Liquid water has 5 times the capacity to absorb heat and produce same temperature change as same mass of sand

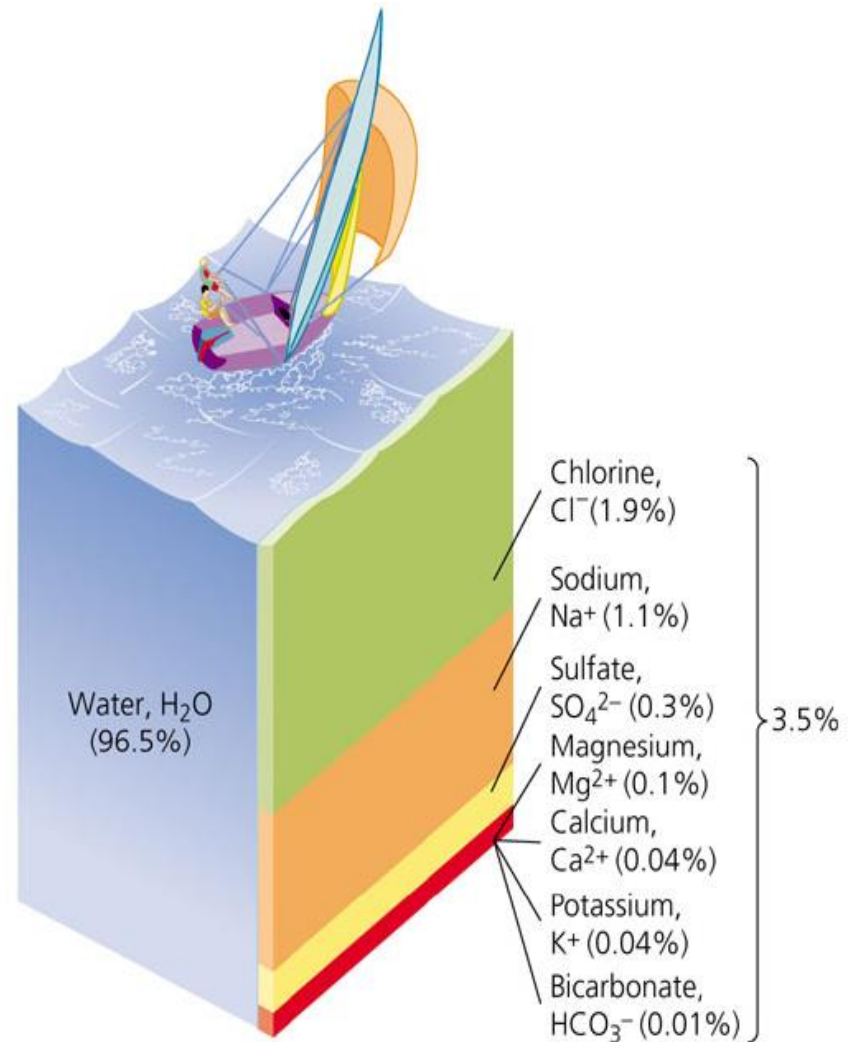
3. Seawater temperature/salinity/density

- Density of seawater is function of **temperature and salinity**
- The temperature range in open sea is **-2~30 °C**
- Changes in temperature and salinity are primarily controlled by surface processes
- The **difference in the density of the seawater** is the main factor that drives the deep ocean circulation
- **Pycnocline (密度跃层)** and **thermocline (温度跃层)** often coincide in water column that separate the surface warmer, less dense water with cold dense bottom water



4. Water as a Solvent

- Water is the most popular solvent
- **Salinity:** The total amount of salt dissolved in seawater
- Salinity is expressed as the number of grams of salt left behind when 1000 grams of seawater are evaporated (parts per thousand, ppt, ‰, psu-practical salinity unit)



- The Ocean is in Chemical Equilibrium
 - Ocean salinity nearly constant, the salinity in ocean is around 35 ppt, this can vary by location
 - Rule of constant proportions (定比定律): Major constituents are in constant ratio with each other, even though the total amount of salt varied slightly from place to place
 - Amount of each constituent going in ocean appears to equal to the amount going out

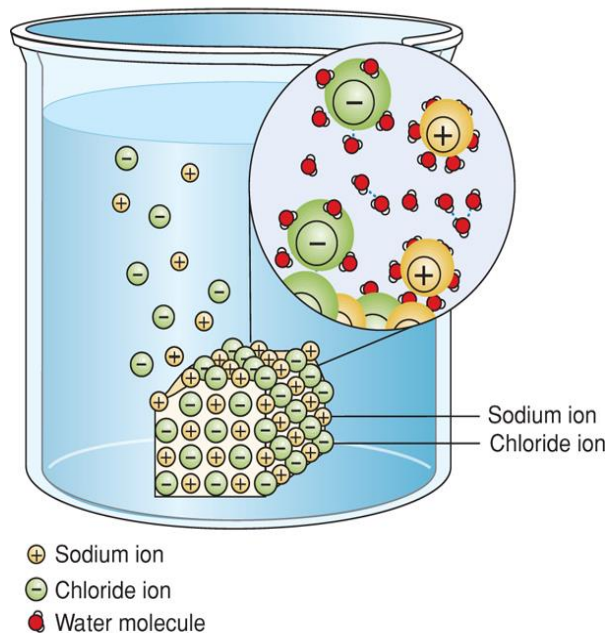


Table 3.1

The Composition of Seawater of 35‰ Salinity

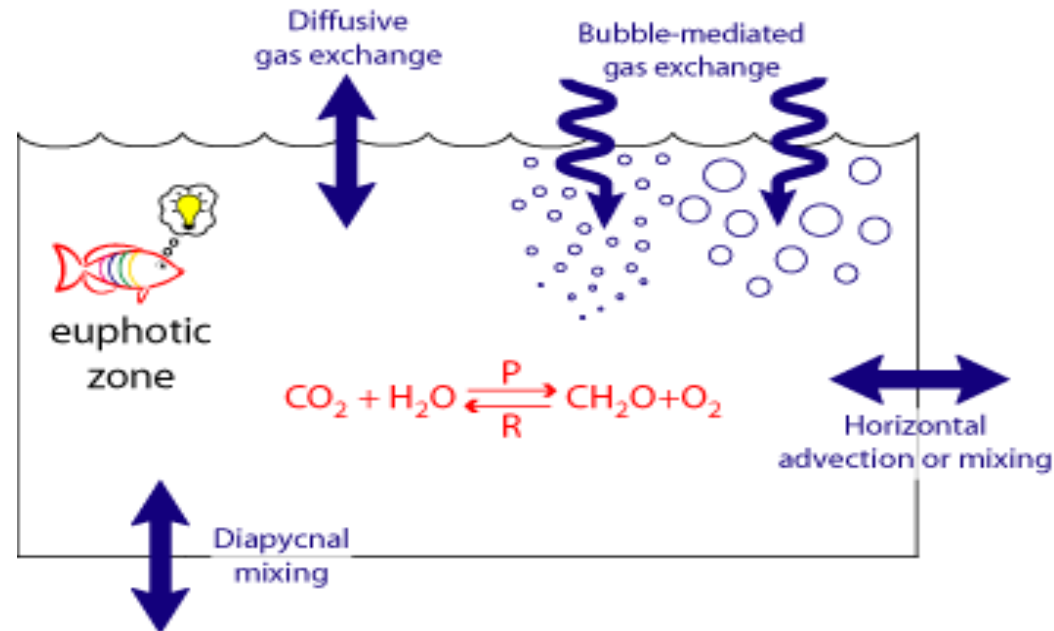
Although the concentration varies slightly from place to place in the ocean, the percentage of total salinity of each ion remains constant.

Ion	Concentration ‰	Percentage of Total Salinity
Chloride (Cl ⁻)	19.345	55.03
Sodium (Na ⁺)	10.752	30.59
Sulfate (SO ₄ ⁻²)	2.701	7.68
Magnesium (Mg ⁺²)	1.295	3.68
Calcium (Ca ⁺²)	0.416	1.18
Potassium (K ⁺)	0.390	1.11
Bicarbonate (HCO ₃ ⁻)	0.145	0.41
Bromide (Br ⁻)	0.066	0.19
Borate (H ₂ BO ₃ ⁻)	0.027	0.08
Strontium (Sr ⁺²)	0.013	0.04
Fluoride (F ⁻)	0.001	0.003
Other dissolved material	<0.001	<0.001

5. Gases in Seawater

- Many gases are dissolved in seawater including:

- Oxygen
- Carbon dioxide
- Nitrogen



- Gases dissolve at the sea surface from the atmosphere. Occasionally, the reverse happens
- Temperature regulates the dissolution of gases: cold water holds gases better than warm water
- Like other aerobic organisms, many organisms in the ocean utilize oxygen and expel carbon dioxide

5.1 O₂ in the Seawater

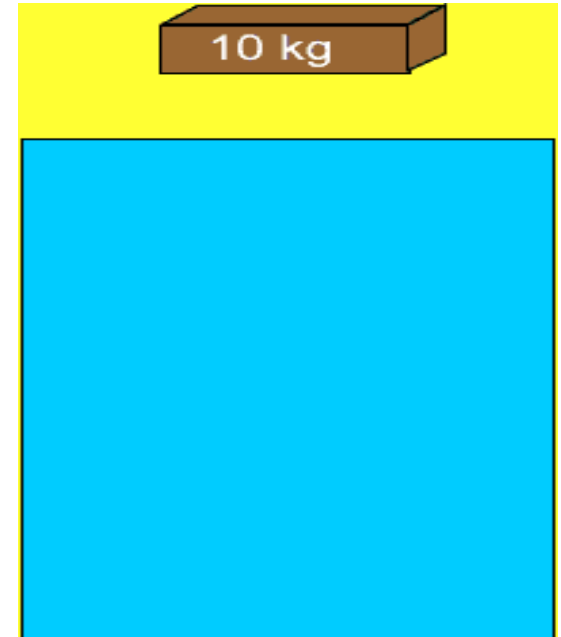
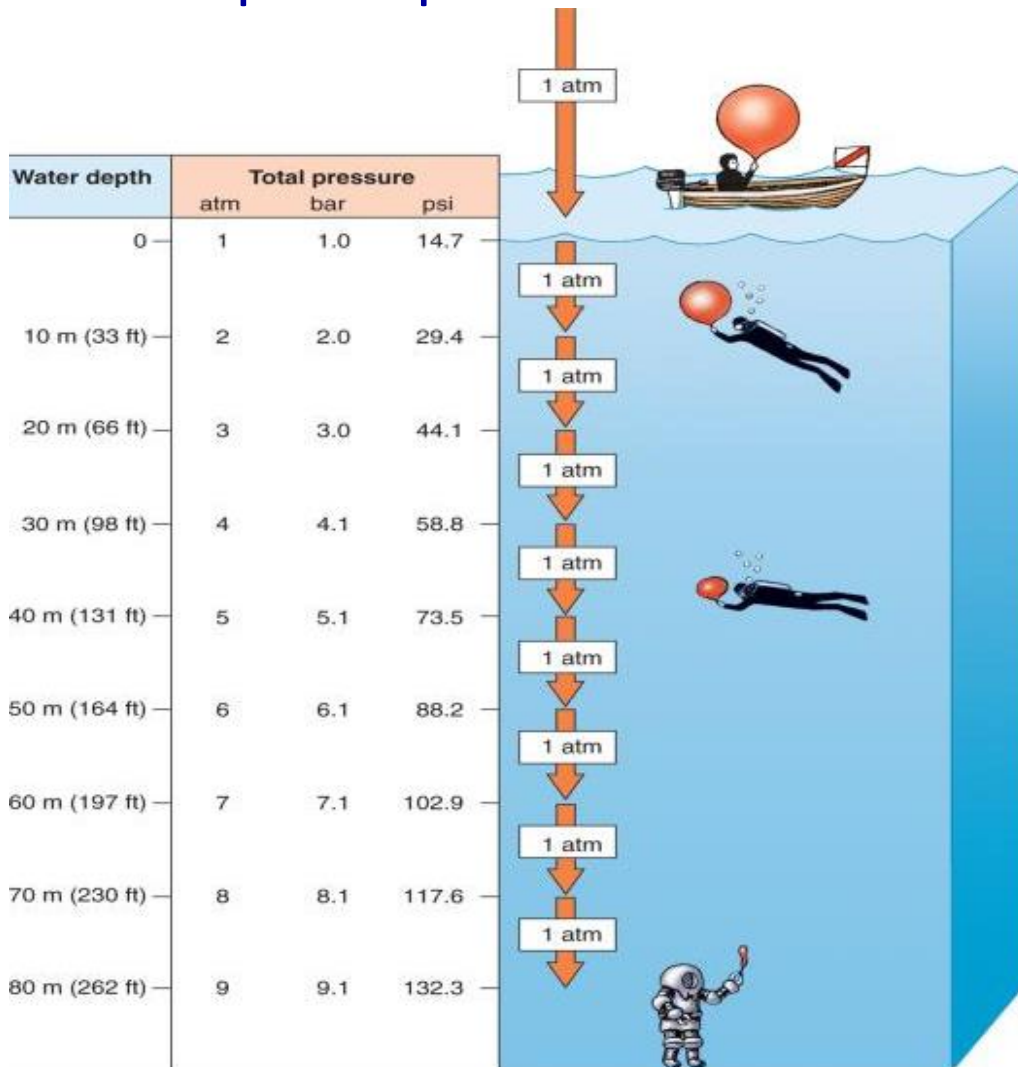
- **Oxygen (O₂) concentration is about 0-8 mg/L** (compare to 210 mg/L in air, or 21% of the total volume), the top 20 m contains the highest concentration of O₂
- **The ocean's oxygen minimum layer**, or the depth at which oxygen becomes depleted, **is usually around 500 m**
- The amount of oxygen in the water is affected by **temperature** and organisms activities (**photosynthesis and respiration**)

5.2 CO₂ in Seawater

- Carbon dioxide (CO₂) is much more soluble than O₂ because it reacts chemically when it dissolves
- CO₂ makes up **more than 80%** of the dissolved gas in the ocean (compared to less than 0.04% of air), and the ocean stores **more than 50 times** as much CO₂ as the atmosphere

6. Pressure

- Pressure increases with increasing depth: increases 1 atmosphere pressure for each 10 m of water increment

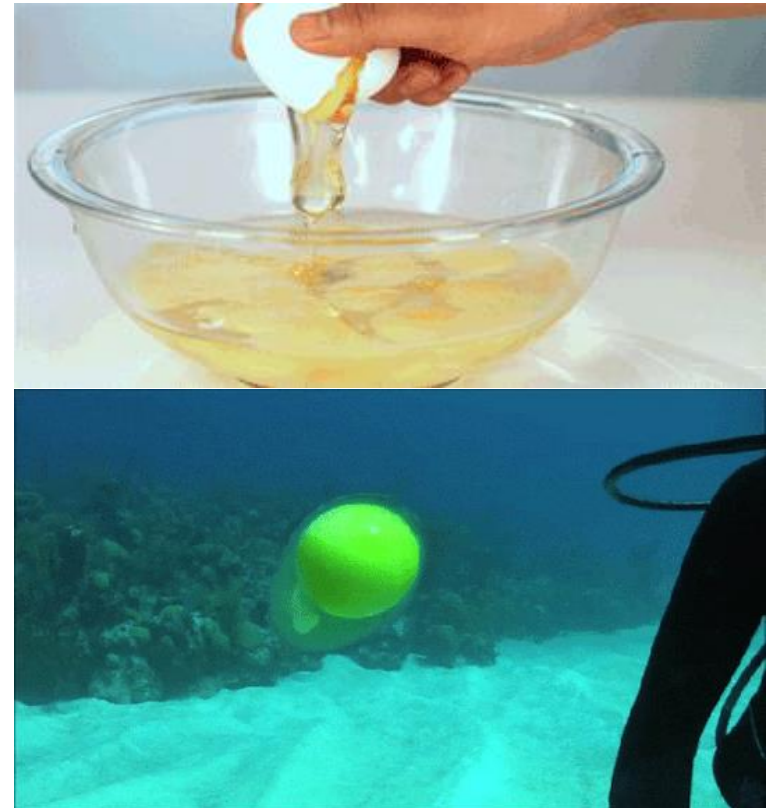


- Pressure poses significant impacts on marine living organisms

Example 1: many fishes, like longfin grouper (玳瑁石斑), have a swim bladder. If the fish is brought to the surface too quickly, the bladder blows so much that it has pushed the stomach out of its mouth

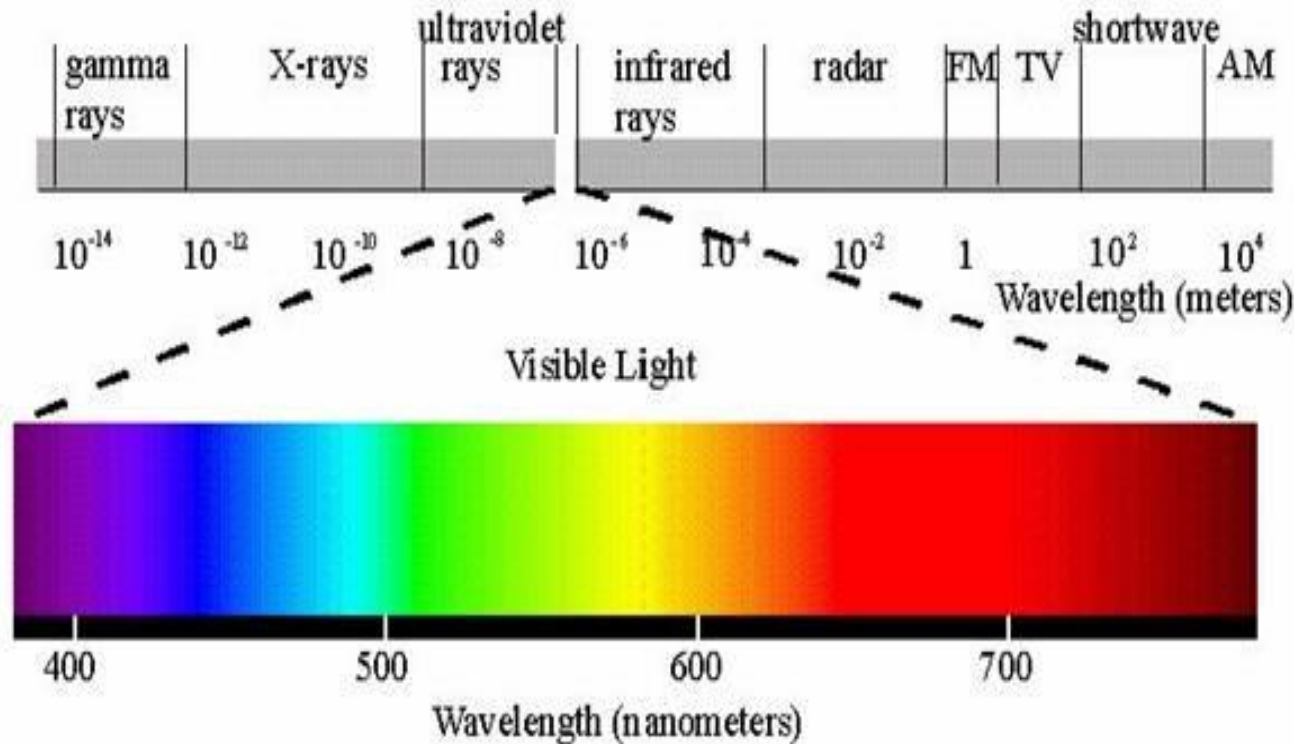


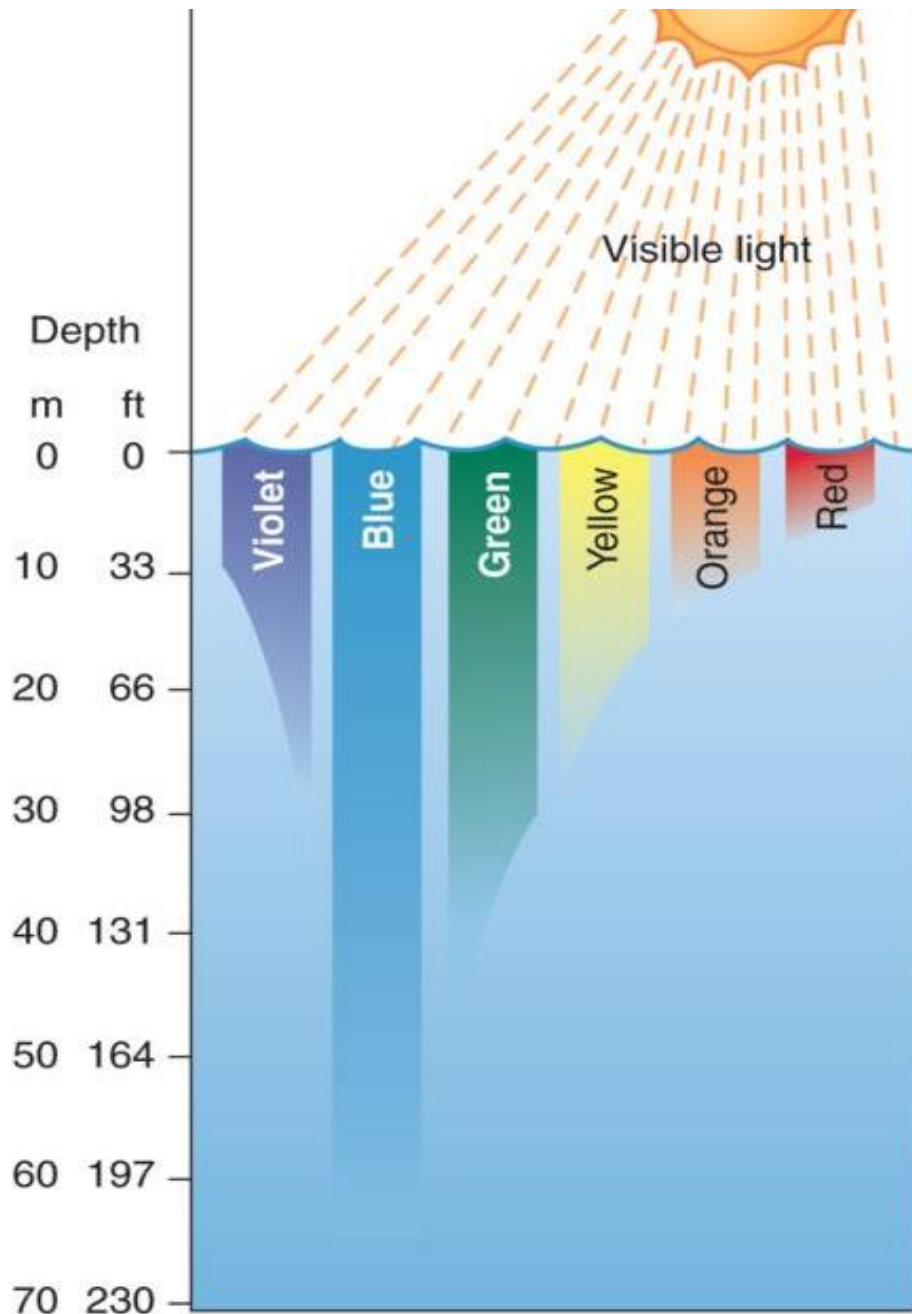
Example 2: Jellyfish contains over 90% water in the body, easier to survive in deep ocean



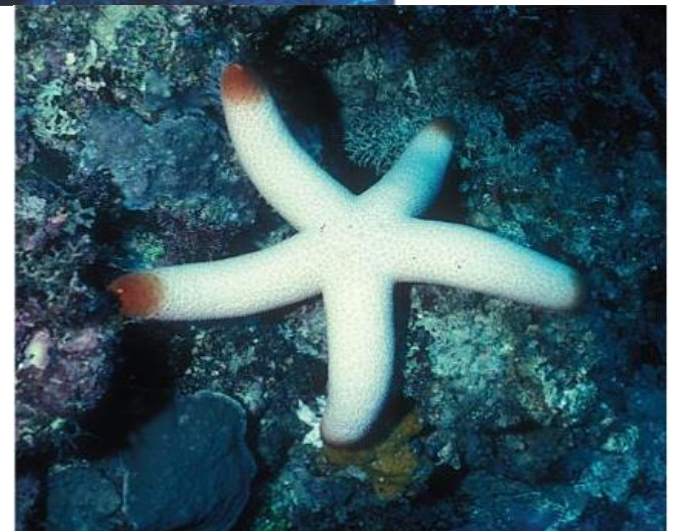
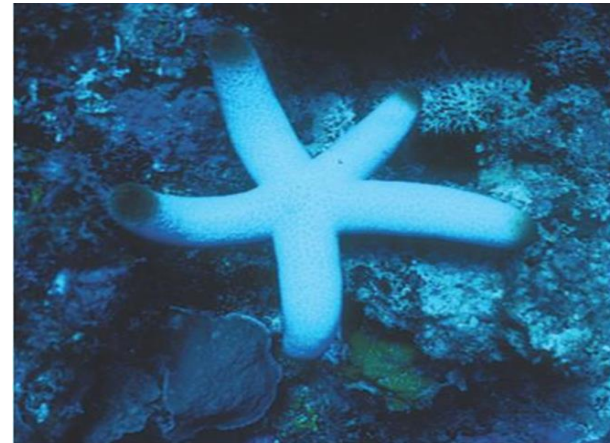
7. Transparency

- Seawater is relatively **transparent**, which is great for photosynthetic organisms living under the sea
- The level of penetration varies greatly depending on the amount of solutes in the water

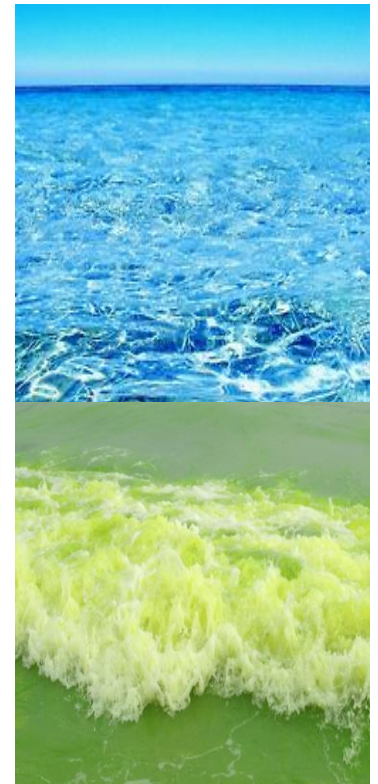
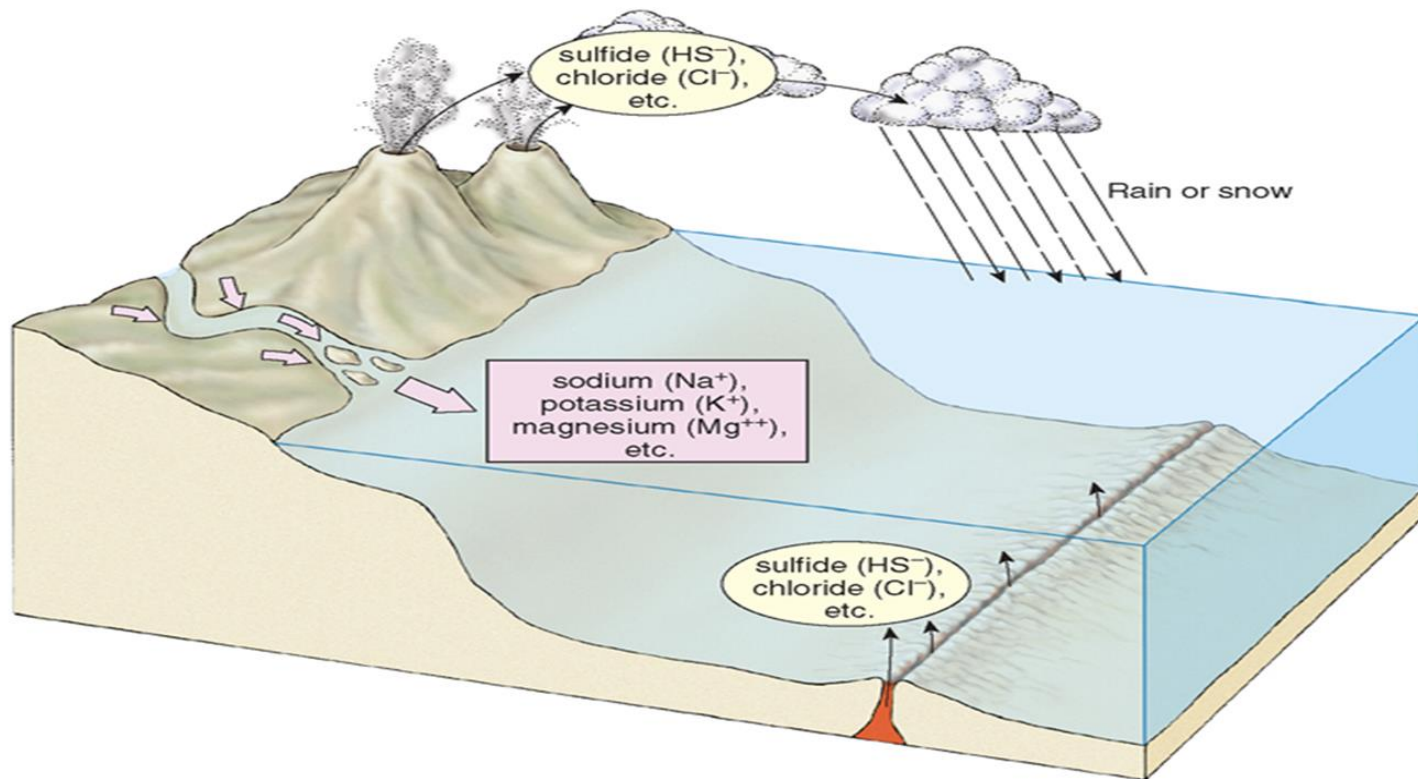




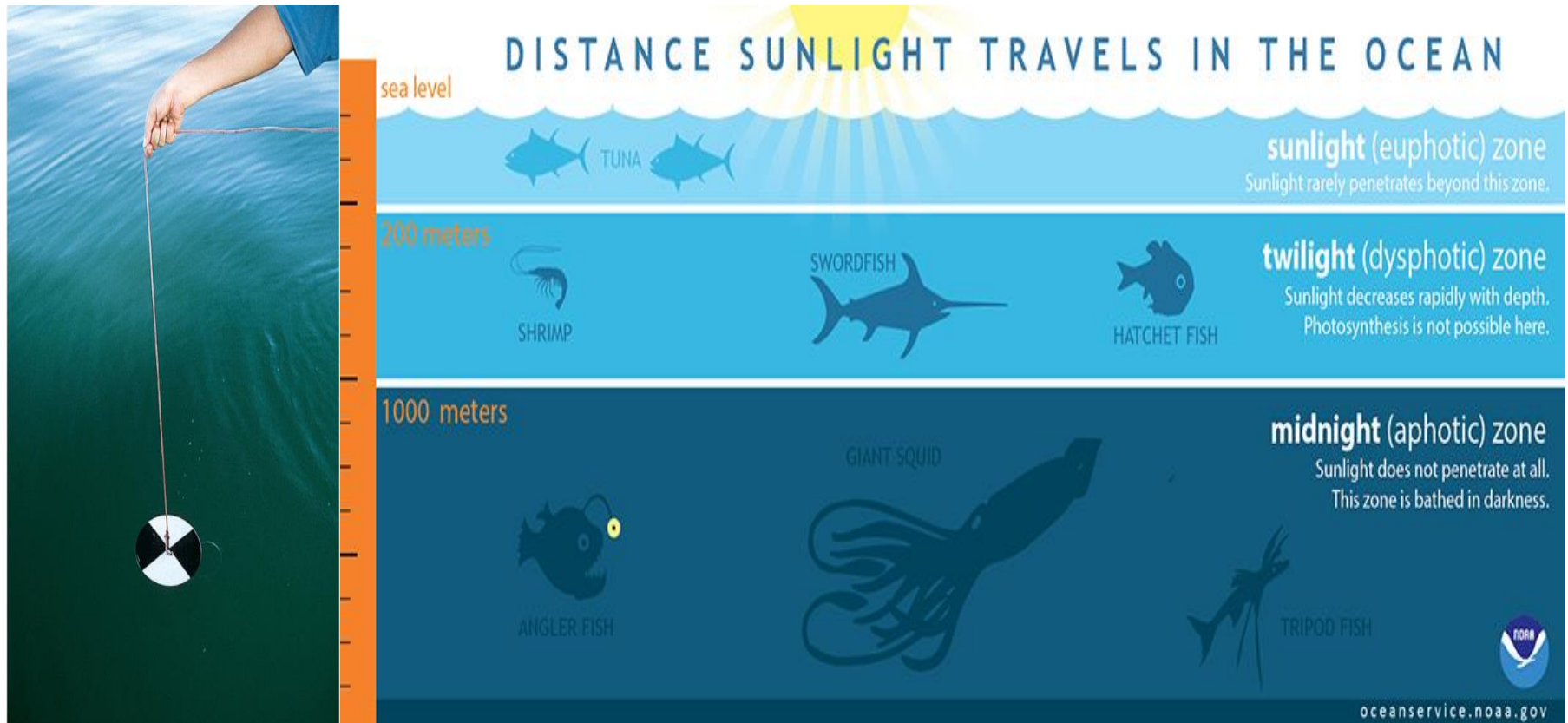
- Water is a very good light absorber of all wavelengths of light except the **blue light**, therefore blue light has the highest penetrating ability



- Changes in ocean color are primarily due to changes in the type and concentration of organisms suspended in the water, or the sediments/solutes in the water from river runoff
 - ❑ Water comes from rivers, from precipitation such as rain and snow, from melting of polar ice
 - ❑ Solutes come from the weathering of rocks, hydrothermal vents and solutes that were delivered in rivers from land run-off



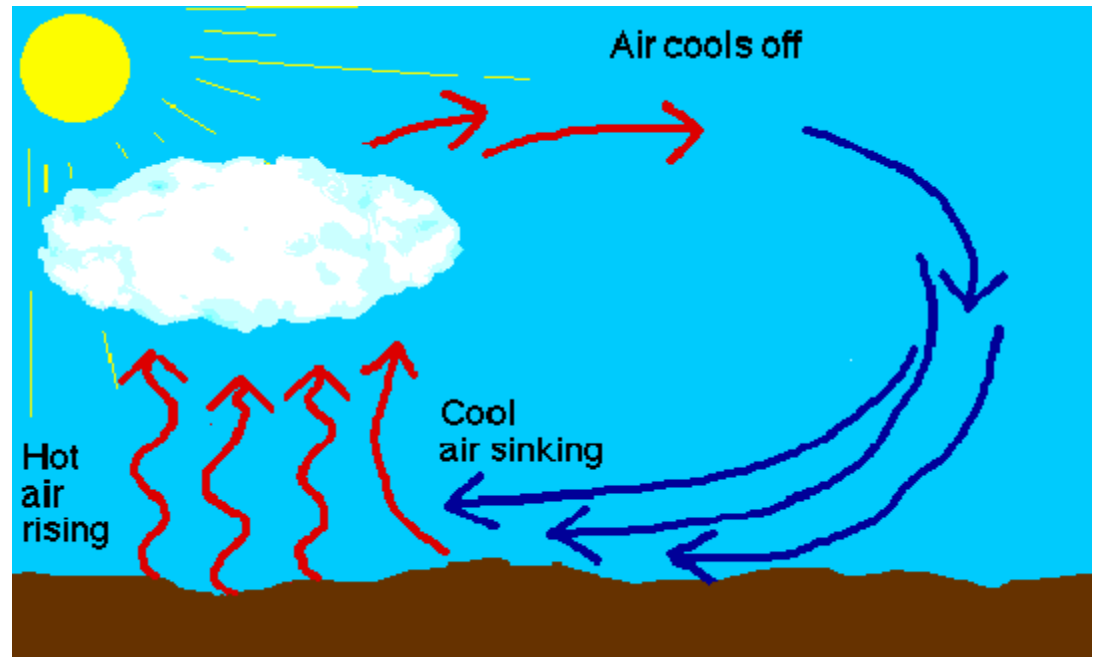
- Transparency, a good indicator of the water quality
- **Secchi dish** is widely used to measure the water clarity
- The light is only available at the first 200 m of surface water, in those more turbulent coastal waters, the depth of euphotic zone (透光层) could be significantly reduced



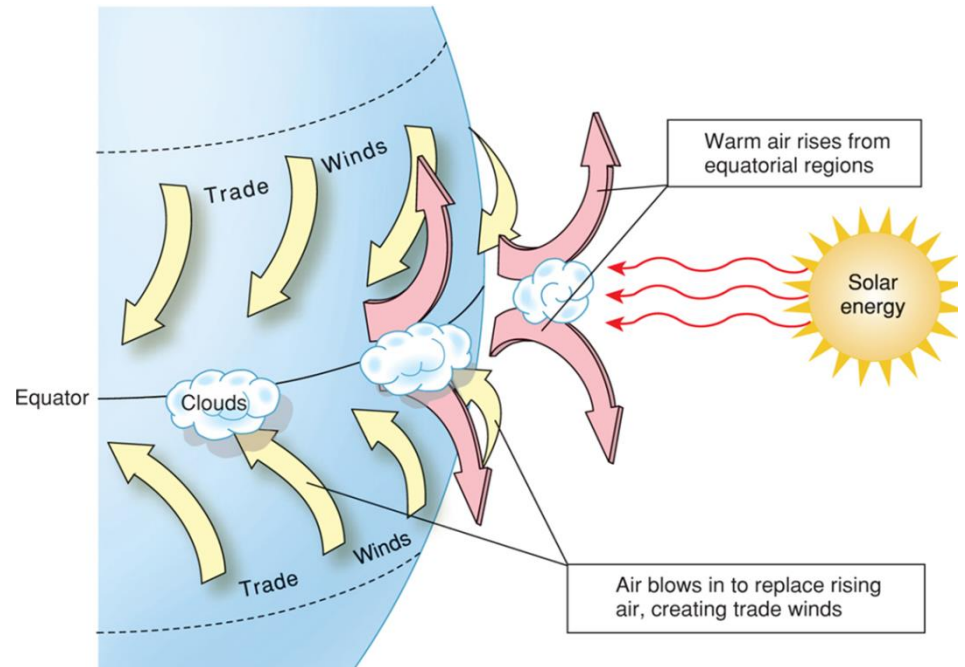
Part 2 Ocean Circulation

- Different physical conditions in the ocean are due to circulation
- Circulation can occur as **waves, tides, currents and gyres**
- Circulation within the ocean is significantly driven by **wind**
- Winds are ultimately driven by **sunlight energy**

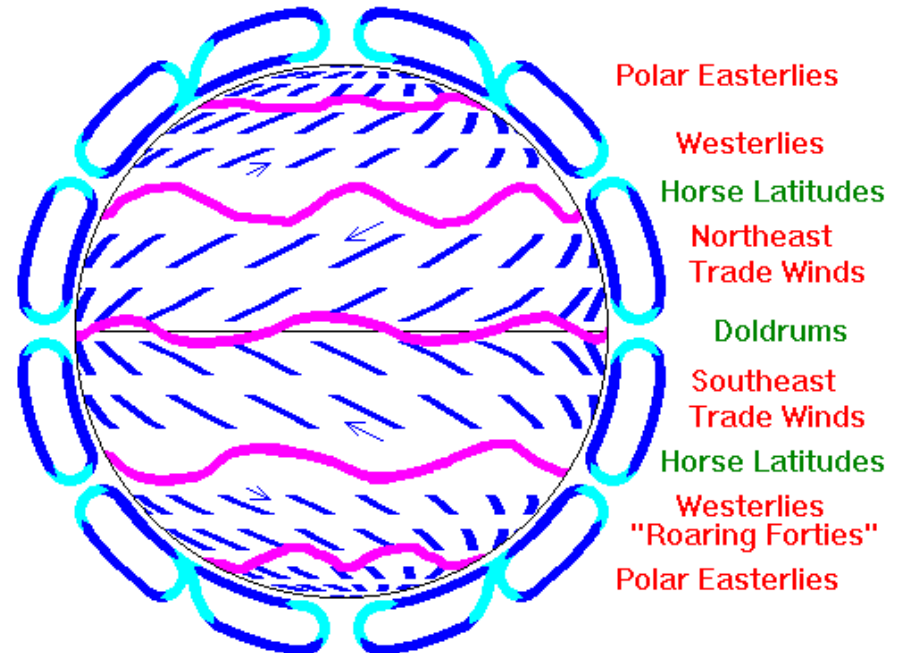
- ❑ As sunlight heats air, air rises
- ❑ Cooler air rushes in to take the place of air that has risen
- ❑ This movement is the source of winds



- Winds created in this manner continuously at the equator are known as the **Trade Winds (信风)**, which is **constant**

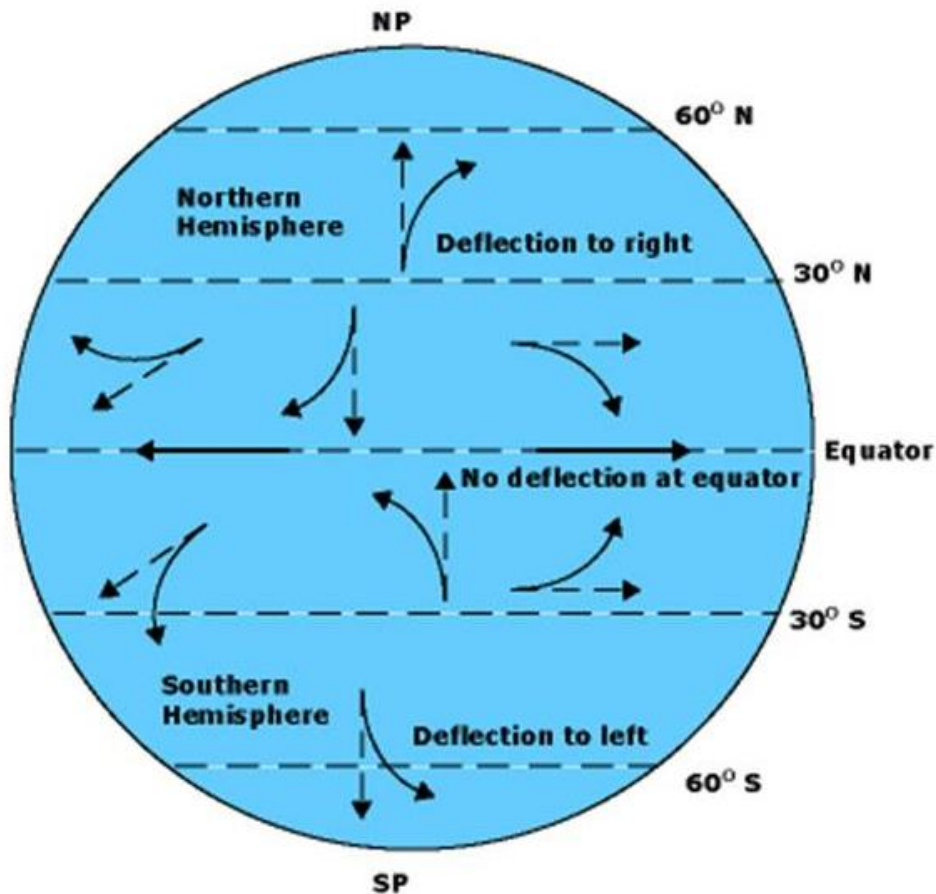


- The Westerlies (西风带) in the mid latitude and the Polar Easterlies (极地东风带) at the poles are less consistent than the Trade Winds



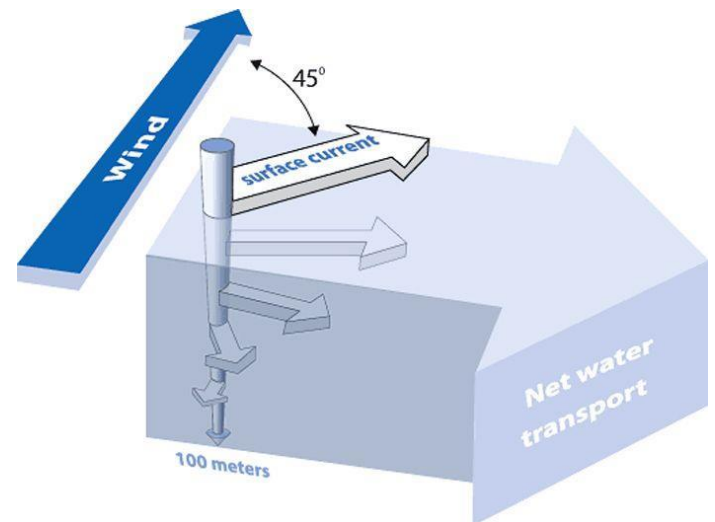
- Coriolis Effect (科里奥效应)

- ❑ Caused by the rotation of the earth
- ❑ Deflects large-scale motions like winds and currents to the right in the Northern Hemisphere and to the left in the Southern Hemisphere



- Ekman Spirals and transport (埃克曼螺旋/输送)

- ❑ The upper-most layer of surface water moves off at an angle of 45 ° because Coriolis effects
- ❑ Lower water layers move at progressively greater angles from the wind in a pattern called an Ekman spiral
- ❑ Ekman transport is the whole Ekman layer moves at 90° from the wind direction



- Surface Currents (表层流)
- ❑ Ocean surface currents are driven by surface winds
- ❑ Surface currents affect upper 10% of the water column
- ❑ The major ocean surface currents in ocean basin are called gyres (5 of them) (大洋涡旋)

